

Shivanand Kumar

SENIOR DATA & APPLIED AI ENGINEER | GENAI & AGENTIC AI | RAG | DATABRICKS

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PROFESSIONAL SUMMARY

Senior Data & Applied AI Engineer with 5 years of experience across HP, Amazon and BYJU'S, combining enterprise Databricks/Spark engineering with production GenAI automation, agentic workflows, retrieval-augmented systems and LLM evaluation. At HP, built LangGraph/Claude Sonnet failure-RCA and automated DataOps workflows, AI-driven data-product automation, lakehouse observability and large-scale production pipelines. Prior Amazon experience includes LLM supervised fine-tuning data quality and prompt-response evaluation alongside production SQL analytics. Independently builds evaluation-driven AI services using Python, FastAPI, Pydantic, Qdrant, structured tools and human-approval controls.

CORE SKILLS

Generative AI & Agentic Systems: LangGraph, LLM orchestration, Claude Sonnet, OpenAI Responses API, function/tool calling, structured outputs, prompt/context design, bounded agent workflows, human-in-the-loop controls

RAG & Retrieval: Retrieval-Augmented Generation (RAG), Qdrant, embeddings, vector retrieval, runbook retrieval, historical-resolution retrieval, evidence grounding, citations, tenant-aware retrieval

AI Evaluation & LLMops: LLM evaluation, versioned evaluation sets, tool/grounding/citation checks, MLflow tracing, token/cost monitoring, guardrails, safety testing, human review

Production AI Engineering: Python, FastAPI, Pydantic, REST APIs, pytest, Docker, GitHub Actions CI, configuration/secrets, structured validation

Data Engineering & Platforms: Advanced SQL, PySpark, Databricks, Spark, Delta Lake, Unity Catalog, ETL/ELT, data modelling, data quality, AWS S3/Glue/Redshift, observability, performance optimisation, BI datasets

EDUCATION

M.Tech, AI & Data Science Engineering (Online) | Indian Institute of Technology (IIT) Patna Jul 2026 - Present

B.E., Mechanical Engineering | UIET, Panjab University, Chandigarh 2016 - 2021

PROFESSIONAL EXPERIENCE

HP Inc. Research & Development Center | Bengaluru

Data Scientist II | Data & AI Platform Engineering Jun 2024 - Present

- Built and productionized the **KLO Agent** across 40 enterprise Databricks jobs, combining system-table enrichment, notebook-level error extraction and a **LangGraph Supervisor** -> **RCA Agent workflow backed by Claude Sonnet**; reduced failure investigation and remediation effort by **~70%**.
- Implemented **retrieval-based reuse of approved historical resolutions**, Unity Catalog persistence, **MLflow token/cost tracing**, Teams notifications and an Upstream Availability Agent that automatically retriggers qualifying downstream jobs through the **Databricks Jobs API**.
- Designed and rolled out an **AI-driven data-product contract automation framework** across **47 Databricks pipelines**, generating and validating configurations, schemas, data sources, target DDLs and **20+ transformations** with repeatable evaluation checks and human approval; reduced documentation effort **from days to hours**.
- Automated monitoring and operational diagnosis across **54 production pipelines**, correlating health and failure signals and reducing recurring manual checks by **85-90%**.
- Led logging deprecation and code standardisation through a reusable migration and refactoring framework; migrated **40 production pipelines** with **zero migration-related runtime failures** and **~90% less manual effort**.
- Optimised firmware-log and Cloudflare telemetry pipelines through partitioning, distributed PySpark processing, join optimisation and caching, delivering **~75% runtime reduction** while lowering compute overhead.
- Led Spark-based deduplication for the Hermes platform, reducing data volume from **21B to 400M rows** while improving downstream performance and infrastructure efficiency.
- Designed lakehouse observability for pipeline health, SLA tracking, proactive monitoring and incident diagnosis, contributing to **~\$1.55M in annual infrastructure savings**.

Amazon Bengaluru & Hyderabad

Business Analyst Jul 2023 - Jun 2024

- Owned analytics for IXD-AMXL supply-chain systems supporting **15M+ users**; translated operational requirements into production SQL pipelines and BI datasets for operational dashboards and leadership decision-making.
- Automated DRR and SSD metric generation and validation using **SQL and Python**, cutting recurring reporting effort by **~70%**.
- Built Dynamic CPT transformations in **SQL**, encoding complex operational rules and reducing runtime by **~30%**.

Recognition: Director-level "Minesweeper" and "Automation Expert" awards for automation impact.

PROFESSIONAL EXPERIENCE - CONTINUED

Amazon

Hyderabad

Machine Learning Data Associate II

Oct 2022 - Jul 2023

- Curated and quality-checked **supervised fine-tuning datasets for LLMs** through annotation, validation and structured dataset-quality workflows.
- Evaluated prompt-response pairs for **relevance, factual consistency, instruction adherence and response quality**; identified recurring failure patterns supporting LLM training-data and evaluation cycles.

BYJU'S

Lucknow

Centre Head - Operations

Oct 2021 - Oct 2022

- Led a **16-member cross-functional team** across centre operations, scheduling, delivery and performance management.
- Built and owned **SQL-driven Student Performance Index dashboards** integrating attendance, assessment scores and tutor metrics; used attendance and performance patterns to guide personalised interventions and achieve a **100% centre rating across North India**.
- Applied data-backed enrolment and operating strategies to drive **~30% quarterly growth** and a record **398+ student registrations**.
- Used operational data insights to support forecasting, regional performance reviews and day-to-day decision-making.

SELECTED AI PROJECTS

Data Platform Reliability Agent | [GitHub](#)

Independent synthetic implementation - no employer code, data or confidential architecture

- Built a **retrieval-augmented reliability agent** with FastAPI, Pydantic, Qdrant, strict read-only tools and tenant-scoped SQL, producing evidence-grounded diagnoses with citations, confidence and approval-required remediation previews.
- Implemented a bounded **OpenAI Responses API tool loop** with deterministic fallback, cross-tenant negative tests and explicit no-autonomous-remediation boundaries.
- Created a **25-case versioned evaluation suite** covering tool selection, incident signature, grounding, citations and approval safety; packaged the project with automated tests, Docker and GitHub Actions CI.

Synthetic Data & Print Recommendation Agent | [GitHub](#)

Independent synthetic implementation

- Built a controlled synthetic-data workflow with **1x/10x/100x generation**, structured recommendations, independent holdout evaluation, diversity and saturation checks, confidence scoring and human-review gates.
- Compared training scales against a separate **12-case holdout**, selecting **10x as the strongest scale** and detecting saturation at 100x instead of treating higher synthetic volume as automatically better.
- Packaged the service with **FastAPI, Pydantic, Docker, automated tests and GitHub Actions CI**, with explicit separation between synthetic labels, holdout evaluation and real-world validation requirements.